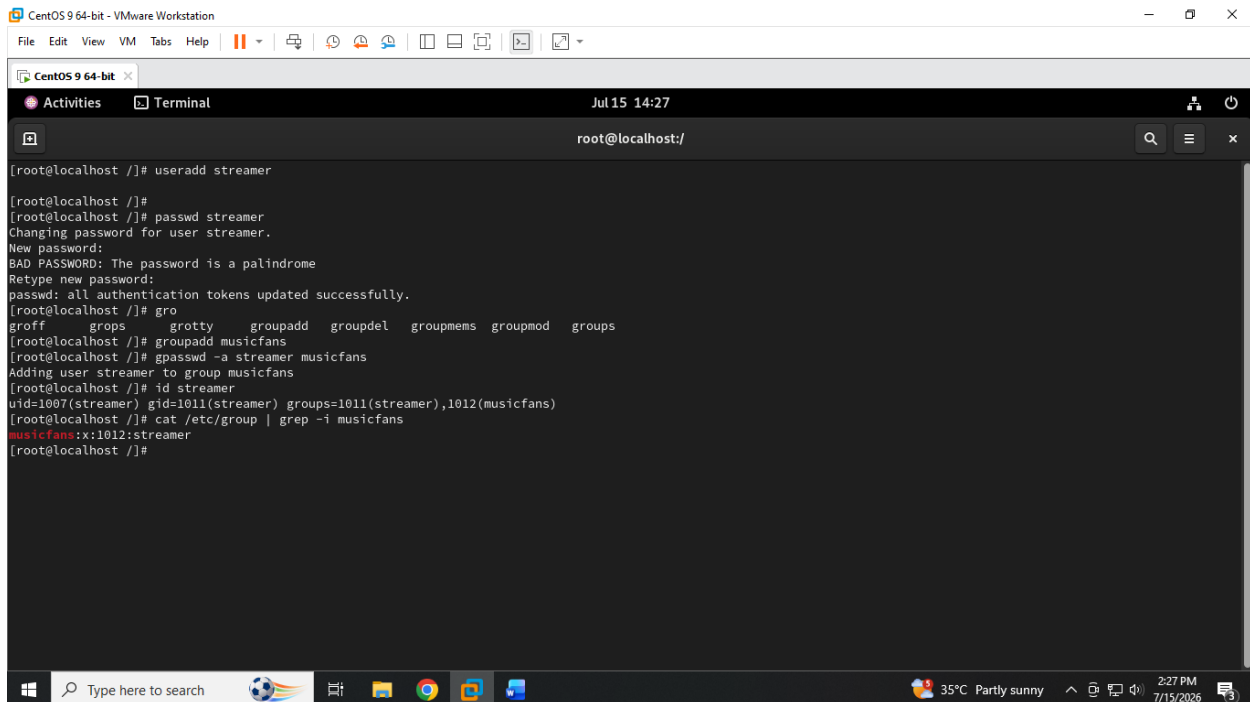


MODULE-7 : MANAGING USERS & SSH SECURITY & UNDERSTANDING OF SELINUX

GUIDED BY : VIPUL MISTRI

Q.1 Create a new user called streamer and a group called musicfans on your Linux system, then add streamer to the musicfans group. Verify the group membership using the id command.

ANS :



```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Jul 15 14:27
root@localhost:/

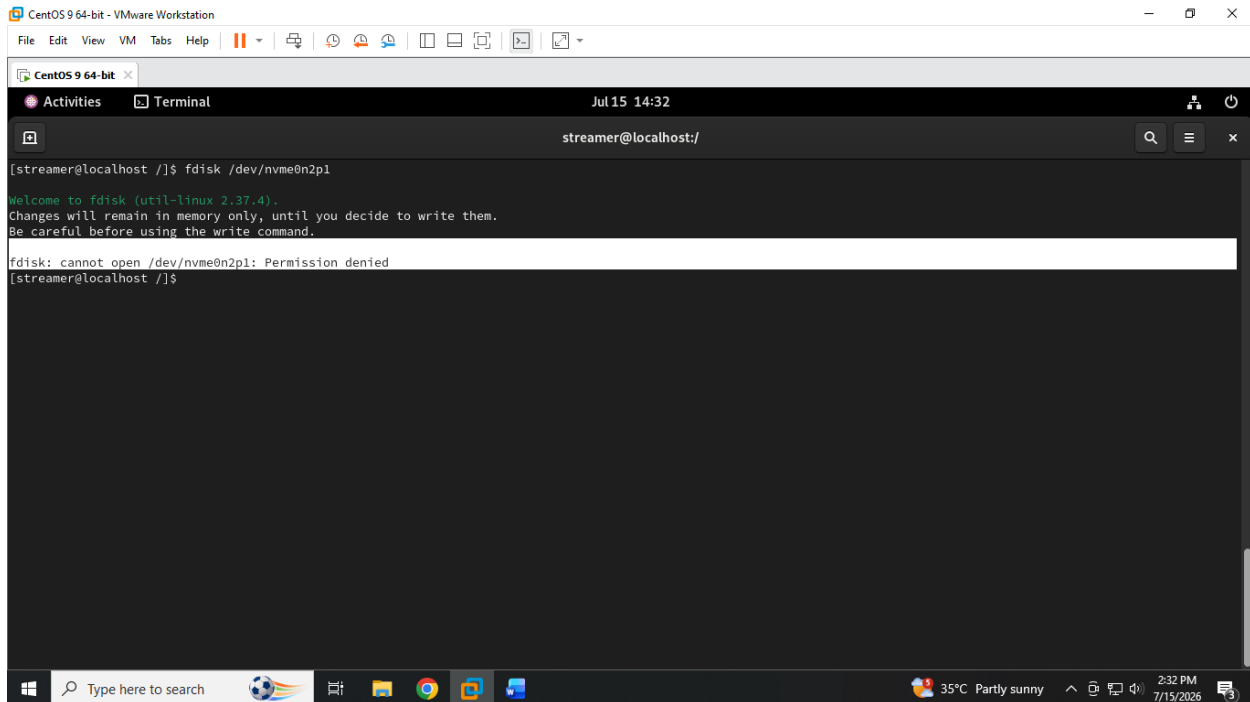
[root@localhost ~]# useradd streamer
[root@localhost ~]#
[root@localhost ~]# passwd streamer
Changing password for user streamer.
New password:
BAD PASSWORD: The password is a palindrome
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost ~]# gro
groff      grobs      grotty     groupadd   groupdel   groupmems  groupmod   groups
[root@localhost ~]# groupadd musicfans
[root@localhost ~]# groupadd -a streamer musicfans
Adding user streamer to group musicfans
[root@localhost ~]# id streamer
uid=1007(streamer) gid=1011(streamer) groups=1011(streamer),1012(musicfans)
[root@localhost ~]# cat /etc/group | grep -i musicfans
musicfans:x:1012:streamer
[root@localhost ~]#
```

MODULE-7 : MANAGING USERS & SSH SECURITY & UNDERSTANDING OF SELINUX

GUIDED BY : VIPUL MISTRI

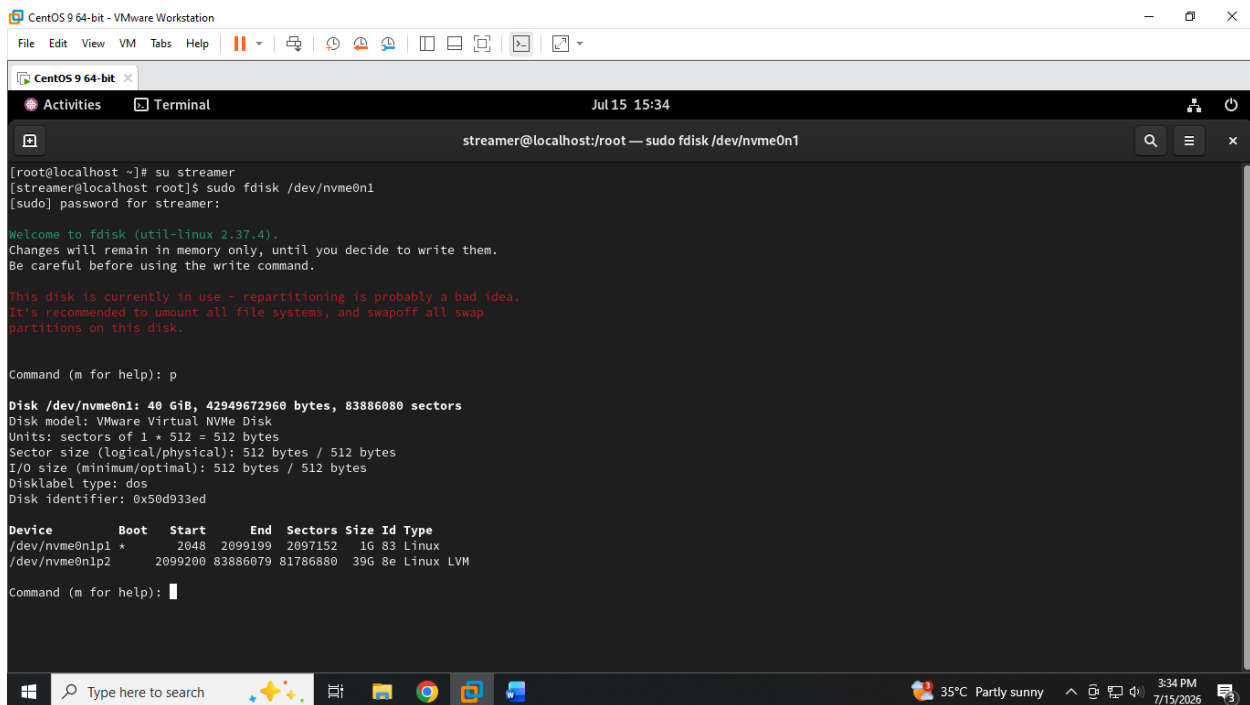
Q.2 Give your current user sudo privileges by editing the /etc/sudoers file using visudo. Test by running a command that requires sudo, such as updating the package list.

ANS :



```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Jul 15 14:32
streamer@localhost:/
[streamer@localhost ~]$ fdisk /dev/nvme0n2p1
Welcome to fdisk (util-linux 2.37.4).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

fdisk: cannot open /dev/nvme0n2p1: Permission denied
[streamer@localhost ~]$
```



```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Jul 15 15:34
streamer@localhost:/root — sudo fdisk /dev/nvme0n1
[root@localhost ~]# su streamer
[streamer@localhost root]$ sudo fdisk /dev/nvme0n1
[sudo] password for streamer:

Welcome to fdisk (util-linux 2.37.4).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

This disk is currently in use - repartitioning is probably a bad idea.
It's recommended to umount all file systems, and swapoff all swap
partitions on this disk.

Command (m for help): p

Disk /dev/nvme0n1: 40 GiB, 42949672960 bytes, 83886080 sectors
Disk model: VMware Virtual NVMe Disk
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x50d933ed

Device      Boot  Start      End  Sectors  Size Id Type
/dev/nvme0n1p1 *    2048    2099199    2097152    1G 83 Linux
/dev/nvme0n1p2      2099200 83886079 81786880    39G 8e Linux LVM

Command (m for help):
```

GUIDED BY : VIPUL MISTRY

ANS :

The image consists of two screenshots of a CentOS 9 64-bit terminal window. The first screenshot shows the installation of openssh-server, enabling sshd, configuring the firewall, and generating an SSH key pair for root. The second screenshot shows the root user logging in via SSH, displaying the SSH banner and the root user's public key fingerprint.

MODULE-7 : MANAGING USERS & SSH SECURITY & UNDERSTANDING OF SELINUX

GUIDED BY : VIPUL MISTRI

```
CentOS version 5 and earlier 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x CentOS version 5 and earl...
Jul 15 3:58 PM
root@localhost:~

inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
    valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:ea:71:0f brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29ea710f
    inet 192.168.17.138/24 brd 192.168.17.255 scope global dynamic noprefixroute ens33
        valid_lft 1507sec preferred_lft 1507sec
    inet6 fe80::20c:29ff:feea:710f/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
root@localhost:~$ ssh-copy-id root@192.168.17.137
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: ssh-add -L
The authenticity of host '192.168.17.137 (192.168.17.137)' can't be established.
ED25519 key fingerprint is SHA256:zjtDMS3addr5nND6ocE4P5VcbuF8w6LQ0EQWicxMzCA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
root@192.168.17.137's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'root@192.168.17.137'"
and check to make sure that only the key(s) you wanted were added.

root@localhost:~$

CentOS version 5 and earlier 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x CentOS version 5 and earl...
Jul 15 3:58 PM
root@localhost:~ - ssh root@192.168.17.137

/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
root@192.168.17.137's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'root@192.168.17.137'"
and check to make sure that only the key(s) you wanted were added.

root@localhost:~$ ssh root@192.168.17.137
Activate the web console with: systemctl enable --now cockpit.socket

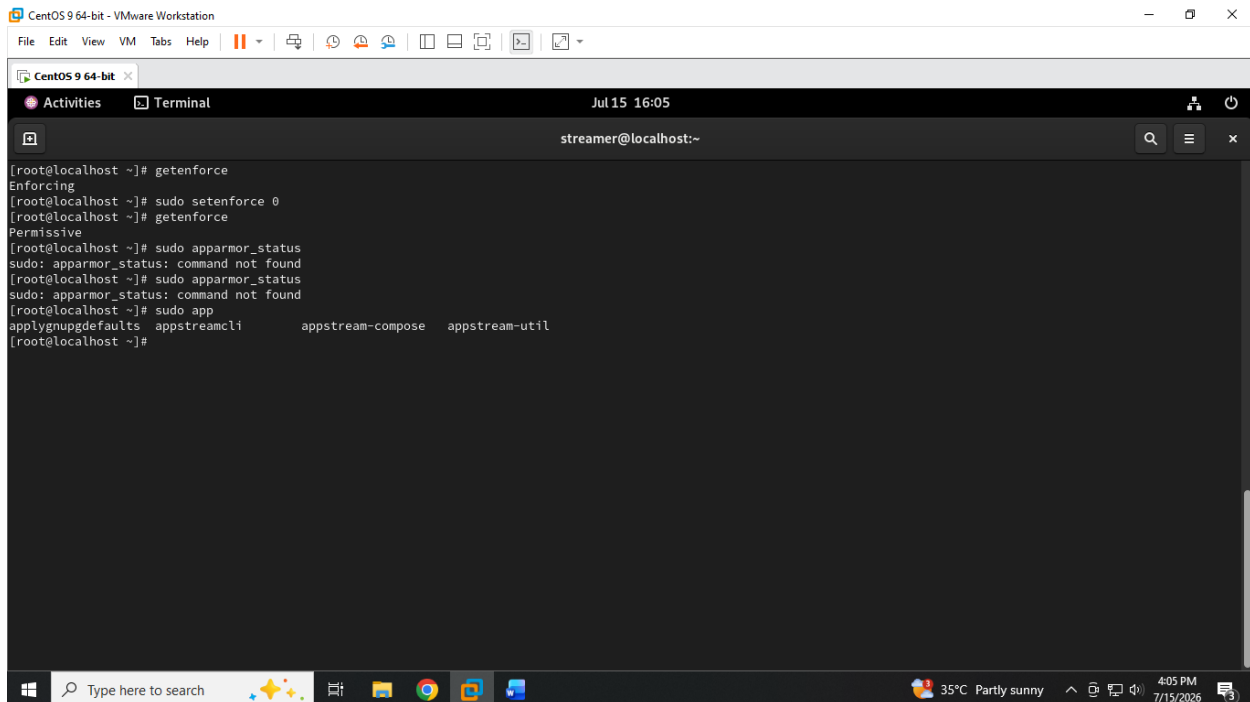
Last login: Wed Jul 15 15:34:30 2026
[root@localhost ~]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:50:9f:03 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.17.137/24 brd 192.168.17.255 scope global dynamic noprefixroute ens160
        valid_lft 1376sec preferred_lft 1376sec
    inet6 fe80::20c:29ff:fe50:9f03/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[root@localhost ~]#
```

MODULE-7 : MANAGING USERS & SSH SECURITY & UNDERSTANDING OF SELINUX

GUIDED BY : VIPUL MISTRI

Q.4 Check if SELinux or AppArmor is enabled on your system. If SELinux is active, use the `getenforce` command to display its current mode, and switch it to permissive mode. If AppArmor is active, list all enforced profiles using `sudo apparmor_status`.

ANS :



```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Jul 15 16:05 streamer@localhost:~
[root@localhost ~]# getenforce
Enforcing
[root@localhost ~]# sudo setenforce 0
[root@localhost ~]# getenforce
Permissive
[root@localhost ~]# sudo apparmor_status
sudo: apparmor_status: command not found
[root@localhost ~]# sudo apparmor_status
sudo: apparmor_status: command not found
[root@localhost ~]# sudo app
applynupgdefaults appstreamcli appstream-compose appstream-util
[root@localhost ~]#
```

MODULE-7 : MANAGING USERS & SSH SECURITY & UNDERSTANDING OF SELINUX

GUIDED BY : VIPUL MISTRI

Q.5 Use ChatGPT to explain the difference between SELinux and AppArmor in one paragraph, then write a summary in your own words and mention one real-world app (like Zomato or Paytm) that would benefit from these security modules.

ANS : **Difference between SELinux and AppArmor (explained by ChatGPT)**

SELinux (Security-Enhanced Linux) and AppArmor are both Linux security modules used to enforce access control policies and protect systems from unauthorized actions. SELinux uses a **label-based security model**, where files, processes, and resources are assigned security labels and access decisions are made according to detailed policies. AppArmor uses a **path-based security model**, where security rules are defined based on the file paths that applications can access. SELinux is generally considered more complex and fine-grained, while AppArmor is often easier to configure and manage. Both help reduce the impact of security vulnerabilities by restricting what applications and users are allowed to do.

Summary in my own words

SELinux and AppArmor are security tools that add an extra layer of protection to Linux systems. They control application permissions so that even if an application is compromised, the attacker has limited ability to access sensitive files or perform harmful actions. SELinux provides stronger and more detailed control but requires more knowledge to manage, while AppArmor provides simpler application-based protection that is easier to set up.

A real-world application such as **Zomato** could benefit from these security modules because its servers handle sensitive information such as user accounts, payment details, and order data. SELinux or AppArmor could help restrict database services, payment systems, and web applications so that a security issue in one component does not compromise the entire system.